
3 Covariance

This is all knowledge required for all of the methods we go through in the course. You will find all of the ‘rules’ for expected values (and thereby covariances) for vectors at <https://www.randomservices.org/random/expect/Matrices.html> (but we are not dealing with ‘Best Linear Predictor’ in this course so you can ignore that section)

3.1 Expected value of variables

What we already know, for variables X , and Y ,

$$V(X) = E \{ [X - E(X)]^2 \} = E(X^2) - \{E(X)\}^2$$

$$Cov(X, Y) = E \{ [X - E(X)][Y - E(Y)] \} = E(XY) - E(X)E(Y) \quad (3.1)$$

and for constants a , and b :

$$V(aX) = E \{ [aX - E(aX)]^2 \} = a^2 E(X^2) - a^2 \{E(X)\}^2 = a^2 V(X)$$

$$Cov(aX, bY) = E \{ [aX - E(aX)][bY - E(bY)] \} = ab [E(XY) - E(X)E(Y)]$$

and

$$\begin{aligned} V(X + Y) &= E \{ [(X + Y) - E(X + Y)]^2 \} \\ &= E \{ (X + Y)^2 - 2(X + Y)E(X + Y) + E(X + Y)^2 \} \\ &= E \left\{ X^2 + 2XY + Y^2 - 2X \underbrace{E(X + Y)}_{=E(X)+E(Y)} - 2YE(X + Y) + E(X + Y)^2 \right\} \\ &= E(X^2) - 2E(X)^2 + E(Y^2) - 2E(Y)^2 \\ &\quad + 2E(XY) - 2E(Y)E(X) - 2E(X)E(Y) \\ &\quad + E(X)^2 + E(Y)^2 + 2E(X)E(Y) \\ &= \underbrace{E(X^2) - E(X)^2}_{V(X)} + \underbrace{E(Y^2) - E(Y)^2}_{V(Y)} \\ &\quad + \underbrace{2 \{ E(XY) - E(X)E(Y) \}}_{2Cov(X,Y)} \\ &= V(X) + V(Y) + 2Cov(X, Y) \end{aligned}$$

For constants a and b ,

$$V(aX + bY) = a^2V(X) + b^2V(Y) + 2abCov(X, Y)$$

3.2 Covariance for vectors

The covariance of two $p \times 1$ vectors $\mathbf{x}$ and $\mathbf{y}$,

$$Cov(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} Cov(X_1, Y_1) & Cov(X_1, Y_2) & \cdots & Cov(X_1, Y_n) \\ Cov(X_2, Y_1) & Cov(X_2, Y_2) & \cdots & Cov(X_2, Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_n, Y_1) & Cov(X_n, Y_2) & \cdots & Cov(X_n, Y_n) \end{bmatrix}$$

From Eq. 3.1, we see that each element can be written $Cov(X_i, Y_j) = E(X_i Y_j) - E(X_i)E(Y_j)$, and consequently

$$\begin{aligned} Cov(\mathbf{x}, \mathbf{y}) &= \underbrace{\begin{bmatrix} E(X_1 Y_1) & E(X_1 Y_2) & \cdots & E(X_1 Y_n) \\ E(X_2 Y_1) & E(X_2 Y_2) & \cdots & E(X_2 Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_n Y_1) & E(X_n Y_2) & \cdots & E(X_n Y_n) \end{bmatrix}}_{E(\mathbf{x}\mathbf{y}^\top)} \\ &\quad - \underbrace{\begin{bmatrix} E(X_1)E(Y_1) & E(X_1)E(Y_2) & \cdots & E(X_1)E(Y_n) \\ E(X_2)E(Y_1) & E(X_2)E(Y_2) & \cdots & E(X_2)E(Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_n)E(Y_1) & E(X_n)E(Y_2) & \cdots & E(X_n)E(Y_n) \end{bmatrix}}_{E(\mathbf{x})E(\mathbf{y})^\top} \end{aligned}$$

To summarise

$$Cov(\mathbf{x}, \mathbf{y}) = E(\mathbf{x}\mathbf{y}^\top) - E(\mathbf{x})E(\mathbf{y})^\top \quad (3.2)$$

In particular, the covariance of a variable with itself, gives the *variance-covariance matrix*

$$V(\mathbf{x}) = Cov(\mathbf{x}, \mathbf{x}) = E(\mathbf{x}\mathbf{x}^\top) - E(\mathbf{x})E(\mathbf{x})^\top \quad (3.3)$$

We leave it as an exercise (in Exercise 4 and also in Lecture 4b), to derive this expression from the definition of $Cov(\mathbf{x}, \mathbf{y})$ in terms of the definition

$$Cov(\mathbf{x}, \mathbf{y}) = E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{y} - E(\mathbf{y}))^\top]$$

3.3 Variance for sums of random vectors

For two $p \times 1$ random vectors $\mathbf{x}$ and $\mathbf{y}$

$$V(\mathbf{x} + \mathbf{y}) = V(\mathbf{x}) + V(\mathbf{y}) + 2\text{Cov}(\mathbf{x}, \mathbf{y}) \quad (3.4)$$

This is easily obtained from

$$\begin{aligned} V(\mathbf{x} + \mathbf{y}) &= E[(\mathbf{x} + \mathbf{y})(\mathbf{x} + \mathbf{y})^\top] - E[\mathbf{x} + \mathbf{y}] E[(\mathbf{x} + \mathbf{y})^\top] \\ &= E[\mathbf{x}\mathbf{x}^\top + \mathbf{x}\mathbf{y}^\top + \mathbf{y}\mathbf{x}^\top + \mathbf{y}\mathbf{y}^\top] \\ &\quad - E(\mathbf{x})E(\mathbf{x})^\top - E(\mathbf{y})E(\mathbf{y})^\top \\ &\quad - E(\mathbf{x})E(\mathbf{y})^\top - E(\mathbf{y})E(\mathbf{x})^\top \\ &= \underbrace{E[\mathbf{x}\mathbf{x}^\top] - E(\mathbf{x})E(\mathbf{x})^\top}_{V(\mathbf{x})} + \underbrace{E[\mathbf{y}\mathbf{y}^\top] - E(\mathbf{y})E(\mathbf{y})^\top}_{V(\mathbf{y})} \\ &\quad + \underbrace{E(\mathbf{x}\mathbf{y}^\top) + E(\mathbf{y}\mathbf{x}^\top) - E(\mathbf{y})E(\mathbf{x})^\top - E(\mathbf{x})E(\mathbf{y})^\top}_{2\text{Cov}(\mathbf{x}, \mathbf{y})} \end{aligned}$$

3.4 Variance and covariance for linear combinations

Let $\mathbf{x}$ be a $n \times 1$ vector of random variables, and $\mathbf{a}$ be a $n \times 1$ vector of constants.

$$V(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}^\top V(\mathbf{x}) \mathbf{a} \quad (3.5)$$

To derive this, use Eq. 3.3, and expectations of inner products from Eq. (2.6)

$$\begin{aligned} V(\mathbf{a}^\top \mathbf{x}) &= E[\underbrace{\mathbf{a}^\top \mathbf{x} (\mathbf{a}^\top \mathbf{x})^\top}_{=\mathbf{x}^\top \mathbf{a}}] - \underbrace{E[\mathbf{a}^\top \mathbf{x}]}_{=\mathbf{a}^\top E(\mathbf{x})} \underbrace{E[\mathbf{a}^\top \mathbf{x}^\top]}_{=(\mathbf{a}^\top E(\mathbf{x}))^\top} \\ &= \mathbf{a}^\top E(\mathbf{x}\mathbf{x}^\top) \mathbf{a} - \mathbf{a}^\top E(\mathbf{x})E(\mathbf{x})^\top \mathbf{a} \\ &= \mathbf{a}^\top [E(\mathbf{x}\mathbf{x}^\top) - E(\mathbf{x})E(\mathbf{x})^\top] \mathbf{a} \\ &= \mathbf{a}^\top V(\mathbf{x}) \mathbf{a} \end{aligned}$$

For an $n \times p$ matrix $\mathbf{A}$ and an $n \times 1$ random vector $\mathbf{x}$

$$V(\mathbf{A}^\top \mathbf{x}) = \mathbf{A}^\top V(\mathbf{x}) \mathbf{A} \quad (3.6)$$

To show this, write $\mathbf{A}$ in terms of its column vectors, as in Eq (2.6), and apply Eq 3.5.

For an $n \times p$ matrix $\mathbf{A}$ and two $n \times 1$ random vectors $\mathbf{x}$ and $\mathbf{y}$

$$V[\mathbf{A}^\top (\mathbf{x} + \mathbf{y})] = \mathbf{A}^\top V(\mathbf{x} + \mathbf{y}) \mathbf{A} \quad (3.7)$$